

Jagadeesh Pulagam

jagadeeshpulagam06@gmail.com — Troy, MI — +1 203 988 4224
linkedin.com/in/jagadeesh-pulagam

Profile Summary

MES Engineer and Microsoft .NET Developer with 2+ years of experience in designing, developing, and deploying applications for manufacturing operations, operations management, RFID-integrated warehouse management systems, and shipping and logistics platforms. Proficient in .NET Core, WPF, SQL Server, and DevOps practices, with a strong background in building high-performance, secure microservices architectures for real-time production monitoring, inventory tracking, and plant floor automation. Solid foundation in **Business Analytics**, skilled in Power BI, and experienced in integrating MES/WMS systems with enterprise platforms. Managed and consumed **RESTful APIs** and worked with **Angular**-based front-end modules for web-based administration and production monitoring tools. Experienced in creating **KPI dashboards** and **visual reports** using Power BI to support data-driven decision-making across manufacturing operations.

Education

University of New Haven, West Haven, CT
Masters in Business Analytics

Jan 2023 – May 2024

Saveetha Engineering College (Anna University), Chennai, India
Bachelor of Engineering in Computer Science

Aug 2018 – Jun 2022

Professional Experience

SmartekPro LLC, Troy, MI MES Engineer / .NET Developer Jan 2024 – Present

- Developed and maintained a custom warehouse management application using **ASP.NET** and SQL Server, integrated with MES systems to manage orders, inventory, and staging workflows.
- Designed and deployed RESTful APIs hosted on **IIS** for pallet registration, asset tracking, and RFID-based inventory movement, including a forklift location tracking system.
- Managed and consumed **RESTful APIs** and worked with **Angular**-based front-end modules for web-based administration and production monitoring tools.
- Built an **Angular**-based admin portal for monitoring shipping orders, pallet positions, floor tags, and order statuses in real time; managed code through **Azure Repos** for version control and team collaboration.
- Integrated **Zebra**, **Alien**, and **Siemens Simatic RF680R** RFID readers to support automated tag reads and validation across warehouse zones.
- Worked with **PLC-driven sensors** and **GPI events** to trigger location-aware workflows and automate floor-level processes.
- Implemented inventory **traceability** for serialized and non-serialized items, capturing full movement from receiving through shipping.
- Configured MES-based **track-and-trace** with genealogy features to record material flow, usage, and process history at each stage.
- Created and optimized SQL Server databases, developing stored procedures and indexing strategies to handle high transaction volumes with consistent performance.

- Defined and managed **MBOM** and **EBOM** structures, including recipe routing for batch processing and MES integration.
- Synchronized MES with master data systems to ensure consistent use of **PlantID**, **WarehouseID**, and **MaterialID** across ERP and production systems.
- Utilized **Inductive Automation Ignition** to configure tags, run diagnostics, and visualize real-time data from the production floor.
- Provided **L1 and L2 support** for MES applications, working closely with end users and the IT team to resolve issues and maintain uninterrupted floor operations.

Tata Consultancy Services (TCS), India .NET Developer Jun 2022 – Jan 2023

- Contributed to backend development of warehouse and enterprise order management applications using **C** and **ASP.NET MVC**, enhancing performance and maintainability.
- Designed and optimized **SQL Server** queries to support data extraction, validation, and reporting across production, inventory, and user activity datasets.
- Built and deployed data visualizations and **Power BI** dashboards for operational metrics and compliance reporting.
- Participated in automation of ETL pipelines using **SSIS** to consolidate plant and order-level data from multiple systems.
- Supported the migration of legacy systems to cloud infrastructure using **Azure SQL Database**.
- Created and documented **RESTful APIs** to support ERP integration and manufacturing workflows across production sites.
- Collaborated with business analysts and QA teams on SOC2-compliant documentation and supported UAT cycles for internal application releases.

Key Projects

LivePoint Smart MES & RFID Warehouse Platform

WPF, .NET Core, Azure SQL, Siemens RF680R, Zebra FX7500, GPI Events

- Developed RFID-based MES functionality to support shipment validation, inventory staging, and floor-tag-activated workflows.
- Integrated Siemens and Zebra readers with PLC-driven sensors using GPI signals for real-time location tracking and automation.
- Streamlined data flow between MES and ERP systems using SQL stored procedures and database synonyms.
- Improved tag read accuracy and lock/unlock operations, reducing shipping delays and floor errors.

Rice Mill WMS and Account Management Portal

ASP.NET MVC, SQL Server, Power BI, Azure App Services

- Built a full-stack web application for inventory tracking, goods receipt, supplier billing, and stock movement management.
- Implemented financial modules for account tracking, purchase histories, and cash flow monitoring with OTP-based transaction approvals.
- Designed SQL schema to capture inward paddy, processed rice, and outbound logistics with precise location mapping.

- Developed Power BI dashboards for real-time visibility into procurement, production performance, and transaction summaries.

Fleetcor – Account Management and Impact Analysis (Industry Collaboration)

SQL Server, SSIS, Data Modeling

- Collaborated with Fleetcor, a global business payments company, on an account-level analysis and treatment impact project.
- Designed a relational data model with over twenty entities in SQL Server and processed large volumes of transactional data using advanced SQL.
- Built SSIS-based ETL pipelines to clean, validate, and transform data across multiple business domains.

Data Integration Lab – University of New Haven (Pompea College of Business)

SSIS, Visual Studio, XML, Excel, Flat Files

- Built SSIS packages for data extraction, transformation, and loading across diverse formats including XML, Excel, text, and RDBMS.
- Configured connection managers and transformations to handle schema drift, data cleansing, and consistent load performance.

Technical Skills

Category	Skills
Languages & Frameworks	C#, Python, .NET Core, .NET, ASP.NET, MVC, Entity Framework, JavaScript, TypeScript
Web & API Development	Angular (Basic), RESTful APIs, Swagger/OpenAPI, JSON Schema
MES/WMS Domains	Manufacturing Execution Systems (PinPoint, RockWell FTPS, Azumuta), Warehouse Automation, Real-time Data Management
Databases	SQL Server, Azure SQL, MySQL, T-SQL, SSIS
Cloud & Hosting	Microsoft Azure, IIS Hosting, Virtual Machines
DevOps & Tools	Azure DevOps, GitHub, CI/CD Pipelines, SSMS, Visual Studio
Testing & QA	xUnit, Code Reviews
ML/Bi Tools (Basic)	Python (Pandas, Scikit-learn), Jupyter Notebooks, Power BI
Collaboration Tools	MS Teams, Excel, Word, PowerPoint
Development Practices	Agile, Scrum, Sprint Planning